ML in computer vision

What are some of the most exciting applications of ML in computer vision today? Can you give me examples of both industrial and research applications?

Great question! Machine learning has pushed computer vision far beyond classical image processing, and today we see both **industrial applications** (things that are already in production) and **research frontiers** (things still being explored in labs). Here are some of the most exciting ones:

🔹 Industrial Applications (Real-World, Deployed)

1. **Autonomous Vehicles (ADAS & Self-Driving Cars)**
   - **What ML does**: Object detection, lane detection, traffic sign recognition, pedestrian tracking, and sensor fusion.
   - **Example**: Tesla Autopilot, Waymo, and Mobileye systems rely heavily on deep learning for perception.
2. **Medical Imaging and Diagnostics**
   - **What ML does**: Detecting tumors, segmenting organs, classifying diseases in X-rays, MRIs, CT scans, retinal images.
   - **Example**: Google DeepMind's AI detects over 50 eye diseases with performance comparable to specialists.
3. **Quality Control in Manufacturing**
   - **What ML does**: Automated defect detection (scratches, cracks, missing components) on assembly lines.
   - **Example**: Intel and Siemens deploy CV systems for PCB inspection and semiconductor defect detection.
4. **Retail & E-Commerce (Computer Vision in Checkout and Recommendation)**
   - **What ML does**: Shelf monitoring, cashier-less checkout, visual search ("find me similar products").
   - **Example**: Amazon Go stores use CV-based tracking to let customers "grab and go" without checkout lines.
5. **Security & Surveillance**
   - **What ML does**: Person re-identification, anomaly detection in CCTV, face recognition.

- **Example**: Airports using CV for automated passport control (biometrics + tracking).

🔹 Research Applications (Frontier, Emerging)

1. **3D Vision & Scene Understanding**
   - Moving from 2D to 3D perception: reconstructing environments, estimating depth, learning from point clouds.
   - **Example**: Neural Radiance Fields (NeRFs) can create photorealistic 3D scenes from just a few 2D images.
2. **Embodied AI & Robotics**
   - Vision-guided agents that can interact with the physical world (grasping, manipulation, navigation).
   - **Example**: Research robots using CV + reinforcement learning to fold laundry, cook, or assemble objects.
3. **Medical Research – Histopathology & Drug Discovery**
   - AI is being trained to analyze cell structures, predict protein folding (AlphaFold), and assist in personalized medicine.
4. **Creative AI (Generative Models for Vision)**
   - Diffusion models (like Stable Diffusion, DALL·E, MidJourney) generating realistic images, video, and 3D content.
   - **Example**: Generating synthetic training data for robotics and simulation.
5. **Multimodal Vision-Language Models**
   - Combining vision + language for tasks like visual question answering, captioning, and reasoning.
   - **Example**: OpenAI's GPT-4o and Google's Gemini can process images and answer questions about them.
6. **AI for Science (Physics, Climate, Biology)**
   - Using CV on satellite images for climate monitoring, analyzing particle collisions in physics, or studying coral reef health.

✅ **Industrial focus** → Reliability, robustness, safety.
✅ **Research focus** → Pushing boundaries in 3D, multimodality, and human-like perception.

Do you want me to expand more on **applications relevant to robotics and mechatronics** (since that's your field), or keep it broad across industries?

Which ideas would be feasible to work on given that it's a semester long project? Focus on the 3D vision and scene understanding subtopic!

Got it 👍 — focusing on **3D vision & scene understanding** for a semester-long project is a smart choice: it's cutting-edge, connects directly to robotics/mechatronics, and has plenty of open-source tools (Open3D, PyTorch3D, Kaolin, etc.). The key is to pick something **ambitious but scoped** so you can actually deliver results in one semester.

Here are **feasible project ideas** with a balance of research flavor and practicality:

🔹 Feasible Semester-Long 3D Vision Projects

1. 3D Object Reconstruction from Multi-View Images

- **Idea**: Take a set of 2D images (from a smartphone or synthetic data) and reconstruct a 3D point cloud or mesh.
- **Methods**: Use Structure-from-Motion (COLMAP) + Multi-View Stereo, or experiment with NeRF-based reconstruction.
- **Why feasible**: Open-source tools exist, you can start with a pipeline and then improve part of it (e.g., denoising, mesh refinement).
- **Deliverable**: A system that turns a few photos into a decent 3D model.

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2. 3D Scene Segmentation with Point Clouds

- **Idea**: Segment objects (tables, chairs, walls) in indoor LiDAR or RGB-D datasets (ScanNet, SUN RGB-D).
- **Methods**: Train/test a point cloud segmentation model (PointNet++, KPConv, or newer transformer-based models).
- **Why feasible**: Plenty of datasets available + pre-trained models. You can focus on improving accuracy, speed, or handling noisy scans.
- **Deliverable**: A point cloud viewer that highlights segmented objects.

3. 3D Pose Estimation of Objects or Robots

- **Idea**: Estimate the 6D pose (rotation + translation) of an object in clutter from RGB-D or point cloud input.
- **Methods**: Use keypoint-based methods, ICP refinement, or train a pose regression network.
- **Why feasible**: Can be tested on synthetic objects (e.g., YCB dataset). This directly connects to robotics (grasping, manipulation).
- **Deliverable**: A demo where your system takes an input scene and outputs estimated object poses.

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4. 3D Shape Completion from Partial Point Clouds

- **Idea**: Given an incomplete point cloud (like a LiDAR scan that misses occluded parts), predict the full object/scene.
- **Methods**: Train a point cloud completion network (PCN, FoldingNet, GRNet) or experiment with diffusion-based approaches.
- **Why feasible**: Datasets exist (ShapeNet, ModelNet), and you can restrict scope to a few categories (chairs, cars).
- **Deliverable**: Side-by-side results showing input (incomplete) vs. your predicted complete shapes.

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5. Scene Change Detection in 3D

- **Idea**: Compare two 3D scans (point clouds or depth maps) and detect what changed (new object added/removed/moved).
- **Methods**: Use point cloud registration (ICP, FGR) + differencing, or design a network to detect changes.
- **Why feasible**: You can simulate this by capturing a room with/without certain objects.
- **Deliverable**: A system that highlights what changed in 3D space.

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6. NeRF (Neural Radiance Fields) for Simple Scenes

- **Idea**: Implement or fine-tune a NeRF model to reconstruct a small object or scene from a few photos.

- **Methods**: Use PyTorch-based NeRF implementations (e.g., Instant-NGP).
- **Why feasible**: Demos exist; your contribution could be in comparing classic multi-view reconstruction vs. NeRF.
- **Deliverable**: Interactive rendering of a 3D scene from novel viewpoints.

🔹 Scoping Advice

- A semester (≈3–4 months) is **not enough to beat state-of-the-art**, but it's enough to:
  - Build on existing pipelines.
  - Apply them to a new dataset or robotic scenario.
  - Add a novel tweak (e.g., improve robustness, visualization, or efficiency).
- The safest "baseline project" is **point cloud segmentation or registration** (lots of tools, datasets, and clear deliverables).
- The "flashier project" is **NeRF or shape completion**, but they need a bit more GPU and time.

👉 Since you're doing a PhD in **mechatronics + robotics**, the most **aligned ideas** would be:

- **3D pose estimation of objects** (for manipulation).
- **Scene change detection** (useful for robotics perception).
- **Point cloud segmentation** (bridge to robot-environment understanding).

Would you like me to **rank these projects by difficulty vs. impressiveness** (easy → hard, practical → flashy), so you can pick one that fits your time/resources?

Assuming that the topic chosen is 3D objects per part segmentation and joint articulation, what are openly available datasets and where can I find?

Perfect 👌 — **3D part segmentation and joint articulation** is exactly in the research sweet spot right now. The good news is that several **open datasets** already exist, many tied to articulated objects (doors, laptops, cabinets, robots).

Here's a breakdown:

🔹 1. PartNet & PartNet-Mobility

- **What**:
  - **PartNet**: ~26,000 3D models from ShapeNet with **fine-grained part segmentation** (chairs, tables, lamps, etc.).
  - **PartNet-Mobility**: Adds **articulations** (hinges, sliders, etc.) to ~2,000 objects.
- **Why relevant**: Has explicit **part segmentation + articulation annotations**.
- **Link**: PartNet ↗ | PartNet-Mobility ↗
- **Format**: Meshes, part labels, articulation parameters.

🔹 2. Shape2Motion

- **What**: Dataset of ~2,000 3D models with **motion annotations** (sliding, rotation, etc.).
- **Why relevant**: Includes both **part segmentation** and **motion attributes** (axis of rotation, translation range).
- **Link**: Shape2Motion ↗
- **Format**: ShapeNet-based meshes + articulation metadata.

🔹 3. SAPIEN

- **What**: A large-scale synthetic dataset + simulator for **articulated objects**.
- **Why relevant**: Contains diverse household objects with physically simulated **joints** (open/close drawers, rotate doors).
- **Link**: SAPIEN ↗
- **Format**: Meshes, articulation models, simulation-ready.

🔹 4. ScanNet

- **What**: RGB-D video dataset (~2.5M views of indoor scenes) with 3D mesh reconstructions and part/object segmentation.
- **Why relevant**: Great for **real-world scans** with object segmentation. No articulation, but useful for **part segmentation pretraining**.
- **Link**: ScanNet ↗

🔹 5. ArtiBoost / RoboArticulation Datasets (Recent)

- **What**: Newer datasets focusing on **robot interaction with articulated objects**.
- **Example**:
  - **ArtiBoost** (2023): Real + synthetic articulated objects with joint annotations.
  - **RoboArticulation**: Designed for robotic manipulation of articulated objects.
- **Why relevant**: More directly linked to robotics tasks like manipulation.
- **Link**: Papers/Repos (can send you latest GitHub repos if you want).

🔹 6. Other Supporting Datasets

- **ModelNet40 / ShapeNet**: Basic 3D object datasets, many later extended with part segmentation (good for pretraining).
- **ABC Dataset** (Autodesk Boundary Representation Collection): Large CAD models with precise part structures (less about articulation, more about parts).
- **YCB Object Set**: Real 3D scans of household items, used in robotics. Not annotated with joints, but widely used for manipulation benchmarking.

🔹 Recommendation for Your Topic

For **3D object per-part segmentation + articulation learning**, the best datasets are:

1. **PartNet-Mobility** → segmentation + joints.
2. **Shape2Motion** → segmentation + motion type.
3. **SAPIEN** → if you want to **simulate robot interaction** with the articulated parts.

👉 A good semester project could be:

- Train a segmentation model on **PartNet-Mobility**.
- Predict **part-level joints/motion** (hinge, slider).
- Validate articulation by simulating in **SAPIEN**.

Would you like me to **map these datasets to specific baseline models/papers** (so you can see what has already been done, and what gap you could tackle for novelty)?

If I were to take a course on ML in the topic chosen, what would the syllabus look like?

Nice — you're essentially asking: *If I were to design or take a semester-long course on ML for **3D object part segmentation and articulation**, what topics should be included?*

Here's a **syllabus outline** that builds up from fundamentals → 3D data handling → part segmentation → articulation modeling → robotics applications. It's structured like a **14-week semester course** (1 lecture/week + assignments + final project).

📘 Course Title:

Machine Learning for 3D Vision: Part Segmentation and Articulation

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Week 1–2: Foundations

- **Lecture 1**: Introduction to 3D Vision & ML
  - Why 3D data? Difference between 2D vs. 3D ML.
  - Overview of tasks: classification, segmentation, reconstruction, articulation.
  - Datasets: ShapeNet, PartNet, Shape2Motion, SAPIEN.
- **Lecture 2**: Representations of 3D Data
  - Point clouds, meshes, voxels, implicit fields (SDF/TSDF, NeRF).
  - Tradeoffs between representations.
  - Libraries: Open3D, PyTorch3D, Kaolin.

Week 3–4: Core ML Techniques for 3D

- **Lecture 3**: Neural Networks for 3D Data
  - PointNet, PointNet++ (point-based models).
  - 3D CNNs on voxels.
  - Graph Neural Networks (GNNs) for meshes.
- **Lecture 4**: Transformer Architectures for 3D
  - Vision transformers in 3D (e.g., Point Transformer).
  - Multi-modal: combining RGB + depth.

Week 5–6: Part Segmentation

- **Lecture 5**: Supervised Segmentation
  - Point cloud segmentation with PointNet++/KPConv.
  - Loss functions (cross-entropy, IoU, Chamfer distance).
- **Lecture 6**: Weakly/Self-Supervised Segmentation
  - Contrastive learning for 3D (e.g., PointContrast).
  - Label scarcity problem & synthetic-to-real transfer.

Week 7–8: Articulation Modeling

- **Lecture 7**: Articulated Objects in 3D
  - Joint types: revolute, prismatic, fixed.
  - Datasets: PartNet-Mobility, Shape2Motion.
- **Lecture 8**: Learning to Predict Articulations
  - MotionNet, Structure-from-Motion approaches.
  - Predicting kinematic graphs from point clouds.
  - Articulation-aware embeddings.

Week 9–10: Applications in Robotics

- **Lecture 9**: Pose Estimation and Manipulation
  - 6D pose estimation methods (PoseCNN, DenseFusion).
  - Aligning articulated objects to target poses.
- **Lecture 10**: Simulation & Interaction
  - Using **SAPIEN** and PyBullet/Mujoco for interaction.
  - RL + CV: teaching robots to manipulate articulated objects.

Week 11–12: Advanced Topics

- **Lecture 11**: Generative Models for 3D Shapes
  - Shape completion, diffusion models, implicit fields.
  - Novel view synthesis (NeRF, Gaussian splatting).
- **Lecture 12**: Multimodal & Language-Grounded 3D Vision
  - CLIP for 3D objects.
  - Text-guided part segmentation/articulation prediction.

Week 13–14: Student Projects & Wrap-Up

- **Lecture 13**: Student Project Progress Presentations.
- **Lecture 14**: Future Directions in 3D Vision
  - Embodied AI, human-object interactions.
  - Open research problems: generalization, data scarcity, sim2real gap.

Assignments / Projects

- **HW1**: Implement PointNet for simple part segmentation (ShapeNet subset).
- **HW2**: Train a 3D segmentation network on PartNet (chairs, tables).
- **HW3**: Articulation prediction on Shape2Motion dataset.
- **Final Project**: Choose a dataset (PartNet-Mobility / SAPIEN) → train/test a model for **part segmentation + articulation** → optionally demonstrate in simulation with robot manipulation.

✅ By the end, you'd:

- Understand 3D representations + ML methods.
- Implement segmentation + articulation pipelines.
- Work with cutting-edge datasets like **PartNet-Mobility**.
- Have a working project demo linking **vision + robotics**.

Do you want me to draft a **reading list (papers + repos per week)** for this syllabus, so you'd have a concrete roadmap with references?

What are open research questions in computer vision that could be explored this semester long project? Focus on the subtopic of 3D per part segmentation and joint articulation.

Great focus 👍 — if you want a semester-long project in **3D per-part segmentation + joint articulation**, the key is to pick a problem where:

1. There's **data available** (PartNet-Mobility, Shape2Motion, SAPIEN).

2. There are **baseline models** to start from.
3. But there's still **room for novelty** (open questions).

Here are some of the **open research questions** you could realistically explore within one semester:

🔹 Open Research Questions in 3D Part Segmentation + Articulation

1. Generalization to Unseen Categories

- **Problem**: Most models work only on object categories seen during training (e.g., chairs, cabinets).
- **Question**: *Can we learn a representation that generalizes articulation understanding to unseen objects?*
- **Example Project**: Train on PartNet-Mobility furniture → test on kitchen appliances. Evaluate transfer.

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2. Segmentation Without Full Supervision

- **Problem**: Part segmentation labels are expensive to annotate.
- **Question**: *Can we use self-supervised or weakly-supervised learning (e.g., contrastive, generative priors) to segment object parts?*
- **Example Project**: Compare supervised PartNet segmentation vs. self-supervised PointContrast or masked autoencoders.

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3. Linking Geometry to Function

- **Problem**: Current methods segment based on shape, but don't fully capture *function* (e.g., door handle = rotation).
- **Question**: *Can we predict articulation type (revolute vs. prismatic) from geometry alone?*
- **Example Project**: Train a classifier on PartNet-Mobility part geometry → predict joint type.

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4. Joint Parameter Estimation

- **Problem**: Even if we know the joint type, estimating the *joint axis* and *range of motion* is hard.
- **Question**: *Can we infer accurate joint parameters from partial scans or noisy point clouds?*
- **Example Project**: Build a pipeline to predict hinge axes for doors/cabinets from Shape2Motion → validate in SAPIEN simulation.

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5. Articulation in Real-World Scans

- **Problem**: Datasets like PartNet are CAD-based (clean geometry), but real scans (ScanNet, RGB-D sensors) are noisy.
- **Question**: *Can we adapt segmentation + articulation models trained on CAD data to noisy real scans (sim-to-real gap)?*
- **Example Project**: Domain adaptation: CAD-trained segmentation → fine-tune on noisy ScanNet data.

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6. Interaction-Guided Segmentation

- **Problem**: Most datasets only provide static geometry, but in reality, interaction reveals parts (e.g., opening a drawer shows segmentation).
- **Question**: *Can robot interaction (simulated in SAPIEN) help discover functional parts?*
- **Example Project**: Use reinforcement learning to pull/push parts, then refine segmentation labels.

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7. Multimodal Cues for Articulation

- **Problem**: Many models use geometry only, ignoring RGB cues.
- **Question**: *Can we combine point clouds + images (RGB-D) to better segment and predict joints?*
- **Example Project**: Extend a segmentation model with RGB features from ScanNet or PartNet-Mobility.

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8. Learning a Universal Articulation Prior

- **Problem**: Current models are category-specific (chairs, tables).

- **Question**: *Is there a category-agnostic articulation prior (like "things with flat surfaces slide")?*
- **Example Project**: Build a meta-learning pipeline that adapts quickly to new categories with few examples.

🔹 Scope for a Semester Project

Given the time frame, the most feasible **semester-long projects** would be:

- **Segmentation without full supervision (Q2)** → practical, doable with existing datasets.
- **Joint parameter estimation (Q4)** → highly relevant for robotics.
- **CAD-to-real adaptation (Q5)** → clear baseline + impactful.

If you want a **flashier but riskier** project:

- **Interaction-guided segmentation (Q6)** → needs simulation (SAPIEN/Isaac Gym).
- **Universal articulation prior (Q8)** → harder, but exciting research.

⚡ So, you could frame your project proposal like:

> *"We aim to explore how to segment 3D object parts and infer articulation properties from partial or noisy observations. Specifically, we will evaluate [X approach] on [PartNet-Mobility / Shape2Motion] and extend it to [open research question]."*

Do you want me to **draft 2–3 concrete project proposals** (with dataset, method, novelty, deliverables) that you could pitch as your semester project?

Suggest me some recent research papers regarding this topic that I can build on during my project. To help you with the search for the papers, I am already reading this paper "Neural Implicit Representation for Building Digital Twins of Unknown Articulated Objects", could you give similar or related papers.

Awesome—since you're already reading **"Neural Implicit Representation for Building Digital Twins of Unknown Articulated Objects" (CVPR'24)**, here's a tightly-curated, recent reading list around **3D per-part segmentation + joint articulation** you can build on. I grouped papers by theme and added quick "why it's useful / build ideas." Citations link to arXiv/CVF/etc.

Core "digital twin" & implicit/field methods

- **Neural Implicit Representation for Building Digital Twins of Unknown Articulated Objects (CVPR'24)**
  Two RGB-D scans → per-part shapes + joint model. Great template for your project's pipeline. arXiv CVF Open Access NV Labs CVPR
- **Learning Implicit Representation for Reconstructing Moving Articulated Objects (ICLR'24)**
  Category-agnostic implicit model with physical constraints; estimates motion + articulation without category skeletons. Good for generalization. arXiv Open Review
- **A-SDF: Learning Disentangled Signed Distance Functions for Articulated Shape Representation (ICCV'21)**
  Classic: disentangles shape vs. articulation; still a strong baseline/regularizer when mixing CAD + real scans. CVF Open Access arXiv

Joint/kinematics estimation (single view / single point cloud)

- **CAPT: Category-level Articulation Estimation from a Single Point Cloud using Transformers (2024)**
  End-to-end transformer for joint type/axis/state from a single cloud—useful if you want minimal input assumptions. arXiv
- **EfficientCAPER (NeurIPS'24)**
  Two-stage, fast part-pose + joint estimation; a good real-time oriented baseline. NeurIPS Proceedings
- **CAPE/CAPER: Toward Real-World Category-Level Articulation Pose Estimation (CVPR'22)**
  Moves beyond fixed kinematics; CAPE-Real benchmark (ReArt). Useful for evaluation protocols. PubMed Liu Liu

Part segmentation + motion jointly

- **Part Segmentation and Motion Estimation for Articulated Objects with Dynamic 3D Gaussians (arXiv'25)**

Jointly infers parts, kinematic tree, and joint poses from point-cloud sequences without point correspondences; leverages dynamic 3D Gaussians—nice bridge to GS methods. arXiv +1 ResearchGate

- **Detection-Based Part-Level Articulated Object Reconstruction from a Single RGB-D (arXiv'25)**
  Cross-category, reconstructs multiple objects with parts + kinematics from one RGB-D frame—good for "single-shot" setups. arXiv

Gaussian splatting & self-supervised articulated modeling

- **ArticulatedGS: Self-supervised Digital Twin Modeling of Articulated Objects using 3D Gaussian Splatting (CVPR'25)**
  Self-supervised, builds per-part digital twins directly in 3D-GS space; promising if you want fewer labels. CVF Open Access

Multimodal / active perception

- **MARS: Multimodal Active Robotic Sensing for Articulated Objects (IJCAI'24)**
  Fuses RGB + point clouds; active strategies to estimate joint parameters—useful for an interaction-guided extension. IJCAI
- **Online Estimation of Articulated Objects with Factor Graphs (2024)**
  Combines vision affordances with interactive kinematic sensing; clean blueprint for "perceive-then-refine" in robotics. University of Edinburgh Research

Category-level articulated pose—foundational

- **Category-Level Articulated Object Pose Estimation (CVPR'20)**
  Introduces ANCSH, estimates per-part 6D pose + joint parameters from a single depth image; classic reference & dataset stats. arXiv CVF Open Access

"Open-vocabulary" / generalization direction

- **Open-Vocabulary 3D Articulated Objects Modeling (arXiv'25)**
  "Articulate AnyMesh"—estimates joint parameters leveraging geometric cues + visual prompting; interesting for language-grounding. arXiv

Surveys / resource compendia

- **Survey on Modeling of Human-made Articulated Objects (CGF'25)**
  Great map of tasks, datasets, and metrics; use to position your contribution.

Wiley Online Library

- **Awesome Articulated Object Understanding (GitHub list, updated 2025)** Curated repo tracking datasets, baselines, and new preprints—handy to keep current. GitHub +1

How to build on these (quick ideas)

- **Your current paper (CVPR'24 Digital Twins) → +GS / self-supervision:** Swap the implicit field stage with **ArticulatedGS** or add a self-supervised loss to reduce reliance on masks/camera extrinsics. CVF Open Access +1
- **Add single-frame priors:** Use **CAPT/EfficientCAPER** to warm-start joint type/axis before your two-state optimization; compare convergence/data needs. arXiv NeurIPS Proceedings
- **Joint part-seg + kinematics from sequences:** Borrow the **dynamic 3D Gaussian** formulation to jointly refine part labels + axes across multiple states, then plug results into your twin builder. arXiv
- **Generalization experiments:** Train/reconstruct on PartNet-Mobility categories, test on appliances (CAPER/ReArt) to report cross-category articulation generalization. PubMed
- **Interaction-guided refinement:** Follow **MARS** or factor-graph online estimation to plan small pokes/pulls in sim (SAPIEN) and update joint posteriors. IJCAI University of Edinburgh Research

If you want, I can pull PDFs/code links for any subset and sketch a **mini-reading plan (2–3 papers/week)** aligned with your semester timeline.

Sources

Based on our discussion, can you draft an abstract for a project proposal regarding the 3D per part segmentation and joints articulation?
Please include motivation, problem definition, methodology, and expected outcomes.

Here's a polished **project proposal abstract** that ties in everything we've discussed (segmentation, articulation, robotics relevance, and feasibility for a semester project):

Abstract

Understanding and modeling the 3D structure and articulation of everyday objects is a fundamental capability for intelligent robotic systems. Robots tasked with interacting in unstructured environments must not only recognize objects but also reason about their constituent parts and the joints that govern their motion. While recent advances in 3D deep learning have enabled progress in part segmentation and shape reconstruction, the challenge of reliably inferring **joint types, parameters, and articulation models** from partial or noisy observations remains largely unsolved, especially in category-agnostic settings.

This project aims to develop a learning-based framework for **3D per-part segmentation and joint articulation estimation** of articulated objects. Building on recent work in neural implicit representations and category-level articulation modeling, we will train and evaluate models using publicly available datasets such as **PartNet-Mobility** and **Shape2Motion**, which provide fine-grained part annotations and joint labels. The methodology will combine a segmentation backbone (e.g., PointNet++ or transformer-based point cloud encoder) with an articulation prediction module to classify joint types (revolute vs. prismatic), estimate joint parameters (axis, range of motion), and link them to segmented object parts. To address real-world challenges, we will also investigate generalization across object categories and robustness to noisy or incomplete point clouds.

The expected outcomes include:

1. A modular pipeline that performs part-level segmentation and predicts corresponding articulation parameters.
2. Quantitative benchmarks on PartNet-Mobility/Shape2Motion, highlighting strengths and limitations of the proposed approach.
3. A demonstration of the method on simulated interaction scenarios in **SAPIEN**, showing how predicted joints enable object manipulation.

By bridging **3D perception** and **functional understanding of articulation**, this work will contribute toward more capable robotic perception systems and lay the groundwork for future digital twin applications.

Would you like me to also **shorten this into a one-paragraph version** (for a course proposal submission form), or keep it in this slightly detailed academic abstract style?

Draft a syllabus for a “Machine Learning for Computer Vision” course with emphasis on 3D per part segmentation and joints articulation sub-topic, including weekly topics, readings, and projects.

Perfect — here’s a full **syllabus draft** for a graduate-level course **“Machine Learning for Computer Vision”** with a **special emphasis on 3D per-part segmentation and joint articulation**.

📘 Course Title

Machine Learning for Computer Vision: From 2D to 3D Articulated Object Understanding

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Course Description

This course introduces core machine learning methods for computer vision, then builds toward advanced topics in **3D scene understanding**, with special focus on **part segmentation** and **articulated object modeling**. Students will learn to represent, process, and reason about 3D data (point clouds, meshes, implicit fields), apply deep learning methods for segmentation and articulation prediction, and implement a semester-long project using state-of-the-art datasets such as **PartNet-Mobility** and **Shape2Motion**.

Course Objectives

By the end of this course, students will be able to:

- Understand machine learning methods for 2D and 3D computer vision.
- Work with 3D data representations and neural architectures (PointNet, Transformers, Implicit Fields).
- Perform 3D **part segmentation** and predict **articulation parameters** (joint type, axis, range of motion).
- Evaluate models on open datasets and extend them to robotic simulation.

- Explore research-level problems like generalization, weak supervision, and sim-to-real transfer.

Prerequisites

- Prior course in machine learning / deep learning.
- Familiarity with Python, PyTorch/TensorFlow.
- Background in linear algebra, probability, and computer vision basics.

📅 Weekly Schedule (14 Weeks)

Weeks 1–2: Foundations

- **Topics**:
  - Intro to ML for CV: supervised, unsupervised, self-supervised learning.
  - From pixels to point clouds: 2D vs 3D vision tasks.
  - Representations of 3D data: voxels, meshes, point clouds, implicit fields.
- **Readings**:
  - *PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation* (Qi et al., CVPR'17).
  - *Survey on Modeling of Human-made Articulated Objects* (CGF'25).
- **Lab**: Intro to PyTorch3D / Open3D.
- **Mini-project**: Visualize ShapeNet objects in different representations.

Weeks 3–4: Core 3D ML Architectures

- **Topics**:
  - PointNet++, KPConv, graph-based methods.
  - Transformers for 3D vision.
  - Loss functions: Chamfer, EMD, IoU.
- **Readings**:
  - *PointNet++: Deep Hierarchical Feature Learning on Point Sets* (Qi et al., NeurIPS'17).
  - *Point Transformer* (Zhao et al., ICCV'21).
- **Lab**: Implement PointNet for toy segmentation task.

Weeks 5–6: Part Segmentation

- **Topics**:
  - Supervised segmentation on PartNet.
  - Weakly/self-supervised segmentation: contrastive and masked modeling.
- **Readings**:
  - *PartNet: A Large-Scale Benchmark for Fine-Grained and Hierarchical Part-Level 3D Object Understanding* (Mo et al., CVPR'19).
  - *PointContrast: Unsupervised Pre-training for 3D Point Cloud Understanding* (Xie et al., ECCV'20).
- **Lab**: Train a segmentation model on chairs/tables.
- **HW1**: Compare supervised vs. contrastive pretraining for part segmentation.

Weeks 7–8: Articulation Modeling

- **Topics**:
  - Kinematics of articulated objects: revolute vs prismatic joints.
  - Predicting joint type/axis from point clouds.
- **Readings**:
  - *Category-Level Articulated Object Pose Estimation* (Wang et al., CVPR'20).
  - *CAPT: Category-level Articulation Estimation from a Single Point Cloud using Transformers* (2024).
- **Lab**: Estimate joint type and axis on Shape2Motion dataset.
- **HW2**: Implement a baseline joint parameter predictor.

Weeks 9–10: Neural Implicit Representations for Articulation

- **Topics**:
  - Signed Distance Functions (SDFs), Occupancy Networks, Neural Radiance Fields.
  - Neural implicit models for articulated objects.
- **Readings**:
  - *A-SDF: Learning Disentangled Signed Distance Functions for Articulated Shape Representation* (Wu et al., ICCV'21).
  - *Neural Implicit Representation for Building Digital Twins of Unknown Articulated Objects* (CVPR'24).

- **Lab**: Use an SDF-based network for shape + articulation learning.

Weeks 11–12: Applications in Robotics

- **Topics**:
  - Sim-to-real challenges in 3D vision.
  - Using SAPIEN / PyBullet for articulated object interaction.
  - Reinforcement learning for manipulation.
- **Readings**:
  - *MARS: Multimodal Active Robotic Sensing for Articulated Objects* (IJCAI'24).
  - *Online Estimation of Articulated Objects with Factor Graphs* (2024).
- **HW3**: Simulate interaction with an articulated cabinet in SAPIEN using predicted joints.

Weeks 13–14: Projects and Future Directions

- **Topics**:
  - Student project presentations.
  - Emerging directions: Gaussian splatting for articulation (ArticulatedGS), open-vocabulary part segmentation, interaction-guided perception.
- **Readings**:
  - *ArticulatedGS: Self-Supervised Digital Twin Modeling of Articulated Objects using 3D Gaussian Splatting* (CVPR'25).
  - *Open-Vocabulary 3D Articulated Objects Modeling (Articulate AnyMesh)* (arXiv'25).
- **Lab**: Wrap-up and final demos.
- **Final Project**: End-to-end pipeline for 3D per-part segmentation + articulation, evaluated on PartNet-Mobility or Shape2Motion, with optional robotics demo.

Grading

- Homework & labs: 40%
- Final project (report + demo): 50%
- Participation/reading discussions: 10%

This syllabus keeps the **general ML-for-CV structure** (intro → representations → models → applications) but keeps the **3D part segmentation & articulation** thread running throughout.

Would you like me to also **suggest a timeline for the final project milestones** (proposal, midterm progress, final demo), so it aligns with this syllabus?

How would this syllabus differ from a general ML course syllabus?

Great question 👍 — the syllabus I drafted is **specialized** to your topic, so it would look quite different from a **general ML course**. Let me highlight the differences clearly:

🔑 Differences Between a General ML Syllabus and This Specialized "ML for CV (3D Articulation)" Syllabus

1. Focus of Content

- **General ML course**:
  - Covers **broad algorithms** (linear regression, logistic regression, decision trees, SVMs, clustering, PCA, neural networks).
  - Applications are diverse (finance, NLP, bioinformatics, recommendation systems, etc.).
- **This syllabus**:
  - Skips many classical ML algorithms in depth, and **dives deep into deep learning for 3D vision.**
  - Everything is tied to **3D per-part segmentation, joint prediction, and articulated objects**.

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2. Data Types

- **General ML course**:
  - Works with tabular data, images (MNIST, CIFAR), maybe text or time series.
- **This syllabus**:

- Entirely focused on **3D data**: point clouds, meshes, implicit fields, RGB-D sequences.
- Datasets: **PartNet-Mobility, Shape2Motion, SAPIEN** → specialized and robotics-oriented.

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3. Algorithms / Architectures

- **General ML course**:
  - Introduces broad ML models (kNN, SVM, Random Forests, CNNs, RNNs, Transformers).
  - Goal: give students breadth across techniques.
- **This syllabus**:
  - Specializes in **3D deep learning architectures**: PointNet/PointNet++, Point Transformers, implicit SDF models, Gaussian splatting.
  - Focus is not on *whether ML works* but on *how ML solves specific 3D CV problems*.

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4. Assignments and Labs

- **General ML course**:
  - Homework = implement logistic regression, train SVM, use scikit-learn on UCI datasets.
  - Maybe one deep learning assignment with MNIST or CIFAR.
- **This syllabus**:
  - Labs = implement 3D segmentation networks, estimate joint axes, simulate articulated objects in SAPIEN.
  - Assignments push students to **experiment with 3D datasets and robotic simulation**.

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5. Final Project

- **General ML course**:
  - Open-ended project in *any* domain: predicting house prices, text sentiment, stock forecasting, etc.
- **This syllabus**:
  - Project must involve **3D per-part segmentation or articulation modeling**.

- Strong robotics/vision grounding → e.g., building a digital twin of a household object.

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6. Reading Material

- **General ML course**:
  - Standard textbooks (e.g., Bishop's *Pattern Recognition and Machine Learning*, Murphy's *Machine Learning: A Probabilistic Perspective*).
  - Classic ML papers (SVM, boosting, dropout, backprop).
- **This syllabus**:
  - Focuses on **recent CVPR/ICCV/NeurIPS papers** about 3D object segmentation and articulation.
  - Students learn by reading and reproducing **cutting-edge research** instead of classical theory.

✅ In short:

- A **general ML syllabus** teaches you *the tools of the trade* across domains.
- This **specialized syllabus** teaches you *how to use ML for a very specific frontier problem*: understanding articulated 3D objects.

Would you like me to create a **side-by-side comparison table (week by week)** between a *general ML course* vs. *this specialized course*, so you can see the shift in emphasis more concretely?